R-LINEAR RIGHT EXACT DIRECT-SUM-COMMUTING FUNCTORS MOD_A → MOD_B ARE EXACTLY MODULES OVER $A \OTIMES_R B$

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original equation 0GNU: sources/stacks-chow/tags.txt, source lines 20804–20804. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GNS: sources/stacks-chow/tags.txt, source lines 20802–20802. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R be a ring. Let A and B be R-algebras. There is an equivalence of categories between*

(1) *the category of R-linear functors $F : Mod_A \to Mod_B$ which are right exact and commute with arbitrary direct sums, and*
(2) *the category $Mod_{A \otimes_R B}$.*

given by sending K to the functor F in (0GNU).

Proof. Let F be an object of the first category. By Lemma 0GNS we may assume $F(M) = M \otimes_A K$ functorially in M for some $A \otimes_{\mathbf{Z}} B$-module K. The R-linearity of F immediately implies that the $A \otimes_{\mathbf{Z}} B$-module structure on K comes from a (unique) $A \otimes_R B$-module structure on K. Thus we see that sending K to F as in (0GNU) is essentially surjective.

To prove that our functor is fully faithful, we have to show that given $A \otimes_R B$-modules K and K' any transformation $t : F \to F'$ between the corresponding functors, comes from a unique $\varphi : K \to K'$. Since $K = F(A)$ and $K' = F'(A)$ we can take φ to be the value $t_A : F(A) \to F'(A)$ of t at A. This maps is $A \otimes_R B$-linear by the definition of the $A \otimes B$-module structure on $F(A)$ and $F'(A)$ given in the proof of Lemma 0GNS. □

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